

CANDIDATE
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BIOLOGY

9700/42

Paper 4 A Level Structured Questions

October/November 2017

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Section A

Answer **all** questions.

Section B

Answer **one** question.

Electronic calculators may be used.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **19** printed pages, **1** blank page and **4** lined pages.

Section A

Answer **all** questions.

- 1 Fig. 1.1 shows a snow leopard, *Panthera uncia*. The habitat of the snow leopard is the high mountains of Central Asia.



Fig. 1.1

Snow leopards usually live and hunt alone. Their main prey is grazing herbivores, such as wild sheep and wild goats. They will also hunt domesticated animals and livestock.

The number of snow leopards has decreased dramatically in the last 40 years.

- (a) (i) Predict the consequences to the ecosystem if the number of snow leopards continues to decrease.

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.....[3]

- (ii) In 2003, the total number of snow leopards was estimated to be between 4080 and 6590.

Suggest **one** reason why the actual number of snow leopards in 2003 may have been higher than the estimated number.

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.....[1]

- (b) The Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) is an agreement between the governments of most countries.

The snow leopard is endangered and is listed on this agreement.

Describe the actions taken by participating governments that help to conserve endangered animals, such as the snow leopard.

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- (c) State how a decrease in the number of snow leopards will affect the genetic diversity of the species.

Outline the possible future consequences of this change in genetic diversity.

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- (d) In 1775 the snow leopard was classified as the only species in the genus *Uncia* and was known as *Uncia uncia*.

More recently, new techniques showed similarities to the tiger, *Panthera tigris*, and so in 2006 the snow leopard was reclassified in the genus *Panthera*.

Suggest the type of evidence used to classify the snow leopard in 1775.

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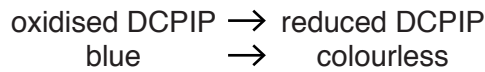
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[Total: 11]

- 2 The light dependent stage of photosynthesis in a suspension of isolated chloroplasts can be investigated using the Hill reaction.

Dichlorophenolindophenol (DCPIP) can be used to follow the process. DCPIP is a blue dye which turns colourless when it is reduced by accepting hydrogen and electrons.



- (a) (i) DCPIP is an artificial hydrogen acceptor that can be used in the Hill reaction.

Name the natural hydrogen acceptor found in chloroplasts that is replaced by DCPIP in the Hill reaction.

.....[1]

- (ii) Outline the way in which hydrogen is made available to reduce the hydrogen acceptor in the light dependent stage of photosynthesis.

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[2]

- (b) A suspension of isolated chloroplasts for measuring the rate of the Hill reaction can be prepared by carrying out the following steps:

- prepare buffer solution with the same water potential as the stroma of chloroplasts
- liquidise (homogenise) spinach leaves in ice cold buffer solution
- filter the liquid and obtain the filtrate
- centrifuge the filtrate to obtain a pellet of chloroplasts
- add the chloroplast pellet to fresh buffer solution in a beaker and mix to obtain a suspension.

Explain the reason for:

- (i) keeping the temperature very low

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[2]

- (ii) using a buffer solution

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- (iii) using a solution of the same water potential as the stroma of chloroplasts.

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- (c) An experiment was carried out to measure the time taken for decolourisation of DCPIP mixed with a suspension of chloroplasts.

The results are shown in Table 2.1.

Table 2.1

replicate	time taken for DCPIP to decolourise/s	rate/s ⁻¹
1	38	
2	43	
3	48	
mean		

Complete Table 2.1 by calculating:

- (i) for the three replicates, the mean time taken for the DCPIP to decolourise [1]

- (ii) the mean rate using the formula:

$$\text{rate} = \frac{1000}{t} \quad \text{where } t = \text{time in seconds.} \quad [1]$$

- (iii) The time taken to decolourise DCPIP was measured at a range of light intensities.

State **and** explain the expected relationship between light intensity and time taken to decolourise DCPIP.

expected relationship

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explanation

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[Total: 15]

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- 3 Fig. 3.1 shows a male red deer, *Cervus elaphus*. Red deer are herbivores, browsing on low vegetation in forests and on waste land.



Fig. 3.1

A history of wild red deer on the western European island of Ireland includes these facts:

- Red deer have lived in Ireland for at least 12 000 years.
- Originally red deer could cross from the neighbouring island of Great Britain to Ireland over a land connection.
- A rise in sea levels at the end of the last Ice Age removed this land connection, separating the red deer on the two islands.
- In the 1800s the number of red deer in Ireland decreased sharply after the main food crop for the human population failed for several consecutive years.
- In the 1900s this decrease in the number of red deer continued as large areas of waste land were drained for agriculture.
- By 1960 red deer were nearly extinct in Ireland, restricted to one population, **A**, of 60 individuals.
- Since then protection has allowed population **A** to increase to over 600 red deer.
- Several new red deer populations, **B**, **C** and **D**, have also been established in different parts of Ireland from individuals brought from Great Britain.

- (a) Use the information given to identify **two** causes of extinction that may have threatened the survival of red deer in Ireland.

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- (b)** Describe how the level of molecular similarity between the red deer in population **A** and population **B** can be investigated.

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- (c)** Explain why some red deer in population **A** show unique molecular features that are not found in any of the red deer of populations **B**, **C** and **D**.

[3]

[Total: 8]

- 4 (a) The body temperature of a human is maintained at its set point of approximately 37°C. If it rises above this temperature, physiological responses begin to return the temperature to its set point. Two of these responses are vasodilation and sweating.

Explain how vasodilation and sweating help to return the body temperature to its set point.

vasodilation

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sweating

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.....[4]

- (b) Diabetes mellitus is a disease where the pancreas is not able to secrete sufficient insulin.

The symptoms of diabetes mellitus include a tendency to drink a lot of water and a loss of body mass.

Suggest why these symptoms occur.

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- (c) A person with diabetes mellitus can use a biosensor to measure the concentration of glucose in their blood.

- (i) Outline how a glucose biosensor works.

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- (ii) Suggest **one** advantage of using a biosensor rather than a dip stick.

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[Total: 12]

5 Huntington's disease is caused by a dominant allele of the gene that codes for the production of the huntingtin protein. This protein affects the development of many different tissues, including brain tissue.

- The Huntington allele contains several repeats of the base sequence CAG, which codes for glutamine.
- This results in a polyglutamine section in the synthesised protein.
- A gene with more than 39 CAG repeats produces a protein that does not fold properly and does not function.
- Symptoms of Huntington's disease usually first appear between the ages of 30 and 45 years.
- There is no treatment for the disease, which is progressive and always fatal.
- Some people with between 27 and 35 CAG repeats do not develop the disease, but may still pass on the Huntington allele to their children, who may develop the disease as the number of repeats tends to increase when gametes are produced.

(a) (i) The amino acid sequences on either side of the polyglutamine section of the huntingtin protein are not changed by the presence of the CAG repeats in the Huntington allele.

Explain why this is so.

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(ii) With reference to the information given, explain why Huntington's disease cannot be treated with gene therapy.

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(b) Young people who have a parent with Huntington's disease can choose to be screened for the presence of the Huntington allele.

(i) State the probability that a young person who has **one** parent with Huntington's disease will inherit the Huntington allele.

.....[1]

(ii) Suggest **one** advantage and **one** disadvantage of screening for Huntington's disease before any symptoms occur.

advantage

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.....

disadvantage

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.....[2]

(c) A couple, in which one partner has the Huntington allele, may choose to use IVF (*in vitro* fertilisation) to have a child.

Any embryos obtained from the IVF procedure can be screened in the following way:

- carry out an embryo biopsy
- use PCR
- test for the presence of the Huntington allele
- only implant embryos that do not contain the Huntington allele.

(i) State what is meant by the term *embryo biopsy*.

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.....[1]

(ii) Explain why PCR is used in this procedure.

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.....[1]

(iii) Outline **two** social or ethical implications of screening embryos in this way.

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[Total: 13]

- 6 Cloves are the aromatic flower buds of the tree *Syzygium aromaticum*. Eugenol is a drug extracted from cloves.

Eugenol affects the movement of sodium ions through the cell surface membranes of sensory neurones.

- (a) Fig. 6.1 shows the effect of eugenol concentration on the percentage decrease in sodium ion movement.

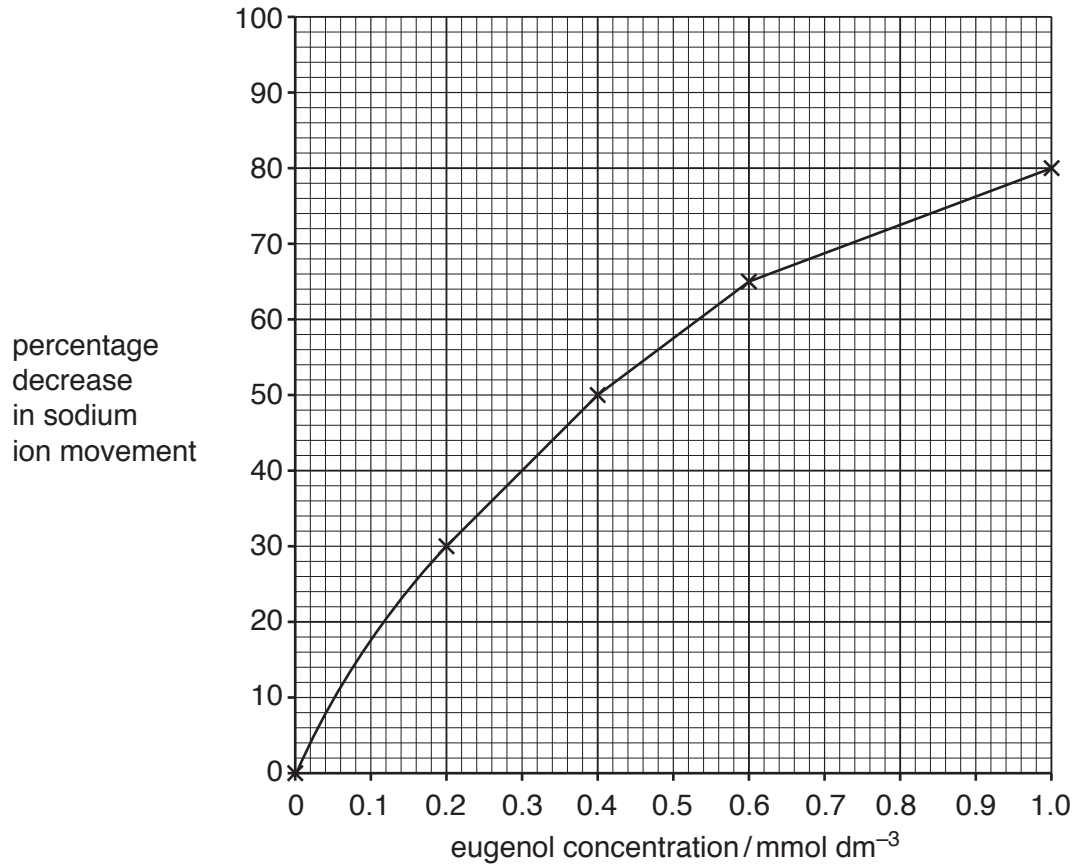


Fig. 6.1

- (i) Describe the effect of eugenol concentration on the percentage decrease in sodium ion movement.

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- (ii) Using Fig. 6.1, determine the percentage decrease in sodium ion movement at a eugenol concentration of 0.5 mmol dm⁻³.

percentage decrease % [1]

- (b)** Eugenol can be used to relieve the pain of toothache by putting it onto the affected tooth and gum.

Suggest **and** explain how the action of eugenol may reduce pain.

.....[5]

[Total: 8]

- 7 (a) In respiration, most ATP is synthesised during oxidative phosphorylation. Some ATP is made by substrate-linked reactions in glycolysis and Krebs cycle.

Describe how ATP is made by substrate-linked reactions.

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- (b) Lipids can be metabolised to provide ATP.

- The enzyme lipase hydrolyses lipids to glycerol and fatty acids.
- The hydrocarbon chain of the fatty acid breaks down into smaller, 2C compounds.
- Each 2C compound reacts with coenzyme A to form acetyl coenzyme A.

- (i) Name the covalent bond in lipids that is hydrolysed by lipase.

.....[1]

- (ii) State the role of acetyl coenzyme A in respiration.

.....[1]

- (iii) Explain why lipids have a higher energy value than carbohydrates.

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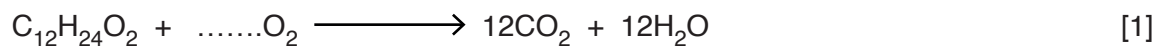
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- (c) The respiratory quotient, RQ, is used to show which substrate is being metabolised by cells. It can be determined using the equation below.

$$\text{RQ} = \frac{\text{molecules of carbon dioxide released}}{\text{molecules of oxygen taken in}}$$

Lauric acid is a saturated fatty acid found in coconuts and has a chain of 12 carbon atoms.

- (i) Complete the equation below which outlines the aerobic respiration of lauric acid.



- (ii) Calculate the RQ value for lauric acid.

Show your working. Give your answer to 2 decimal places.

answer =[2]

[Total: 9]

- 8 (a) The sweet pea, *Lathyrus odoratus*, is a flowering plant that grows in many parts of Europe. The inheritance of flower colour and shape of pollen grains is controlled by genes that display autosomal linkage.

Explain what is meant by *autosomal linkage*.

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.....[2]

- (b) In sweet peas, the allele for purple flowers is dominant to the allele for red flowers and the allele for long pollen grains is dominant to the allele for round pollen grains.

A sweet pea plant that is heterozygous for both purple flowers and long pollen grains is crossed with a sweet pea plant with red flowers and round pollen grains. The results of this cross are shown in Table 8.1.

Table 8.1

offspring phenotype	percentage of offspring
purple flowers, long pollen grains	44
red flowers, round pollen grains	44
purple flowers, round pollen grains	6
red flowers, long pollen grains	6

Describe **and** explain the results shown in Table 8.1.

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- (c) Different alleles affecting flower colour in the sweet pea plant are the result of gene mutations.

State how a gene mutation may occur.

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[Total: 9]

Section B

Answer **one** question.

- 9 (a) Describe how the structure of a mitochondrion is related to its function. [8]
- (b) Describe respiration in anaerobic conditions in mammalian liver cells **and** describe how this differs in yeast cells. [7]

[Total: 15]

- 10 (a) Explain how meiosis can lead to genetic variation. [8]
- (b) Explain how the presence of a mutant allele can result in albinism. [7]

[Total: 15]

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